AWS-VPC

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김현욱

2025-08-22

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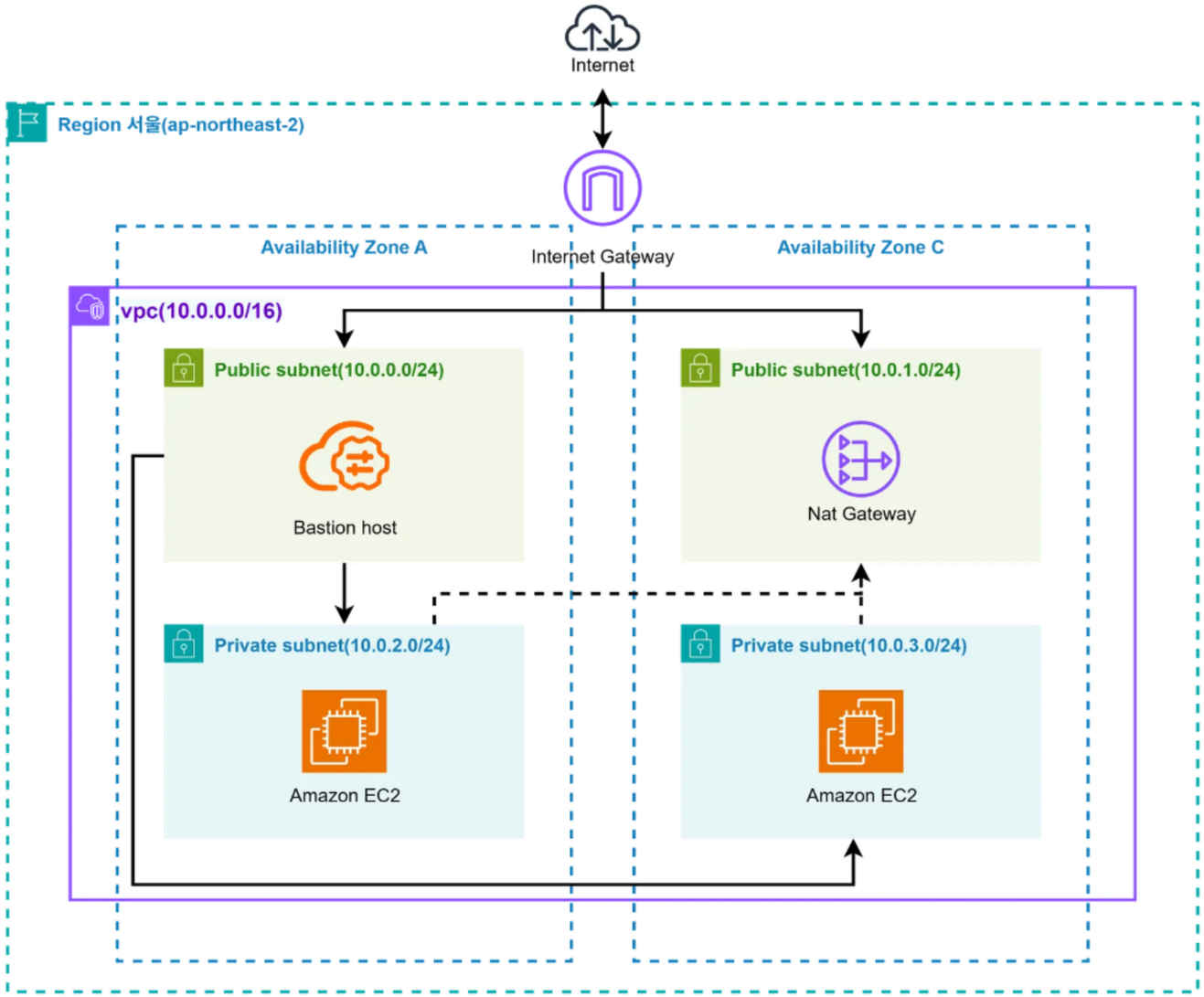
1. 구성도


Internet
Region 서울(ap-northeast-2)
Internet Gateway
Availability Zone A
Availability Zone C
vpc(10.0.0.0/16)
Public subnet(10.0.0.0/24)
Bastion host
Public subnet(10.0.1.0/24)
Nat Gateway
Private subnet(10.0.2.0/24)
Amazon EC2
Private subnet(10.0.3.0/24)
Amazon EC2

2. 주요 기술

2.1 Bastion host

- Bastion host란 Public Subnet에 위치한 EC2 Instance
- 외부에서 직접 접근할 수 없는 Private Subnet에 대신 연결해주는 보안 관문(Proxy) 서버

- 관리자가 VPC 외부에서 Private Subnet 인스턴스에 접근할 방법이 없음
- Bastion Host를 거쳐 안전하게 접속 가능

- 관리자가 Bastion Host에 SSH 연결
- Bastion Host에서 Private Subnet 인스턴스에 SSH 연결

2. 주요 기술

2.2 NAT Gateway

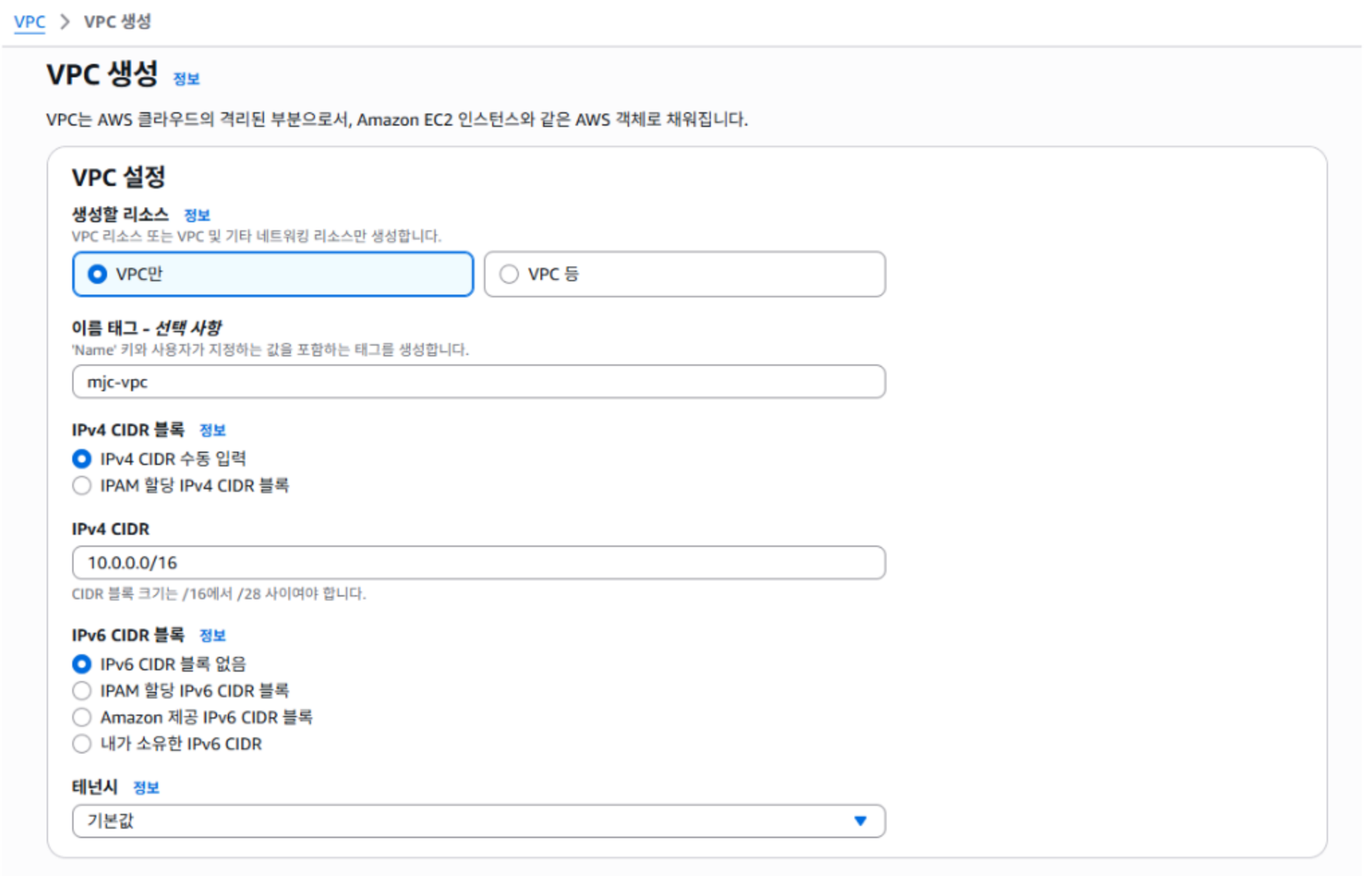
- NAT Gateway란 Private Subnet에서 외부 인터넷으로 통신할 수 있도록 돕는 관문 역할
- Private IP → Public IP 로 변환하여 인터넷과 통신 가능

- Private Subnet 인스턴스가 직접 인터넷에 노출되지 않고도 외부 패키지 설치, 업데이트 등 아웃바운드 통신 가능
- 보안성을 유지하면서 외부 리소스 접근 지원

- Private Subnet 인스턴스가 NAT 게이트웨이를 통해 요청을 보내면, Private IP가 Public IP 로 변환되어 인터넷으로 송신
- 외부 응답은 다시 NAT 게이트웨이를 거쳐 Private Subnet 인스턴스로 전달

3. VPC

3.1 VPC 생성



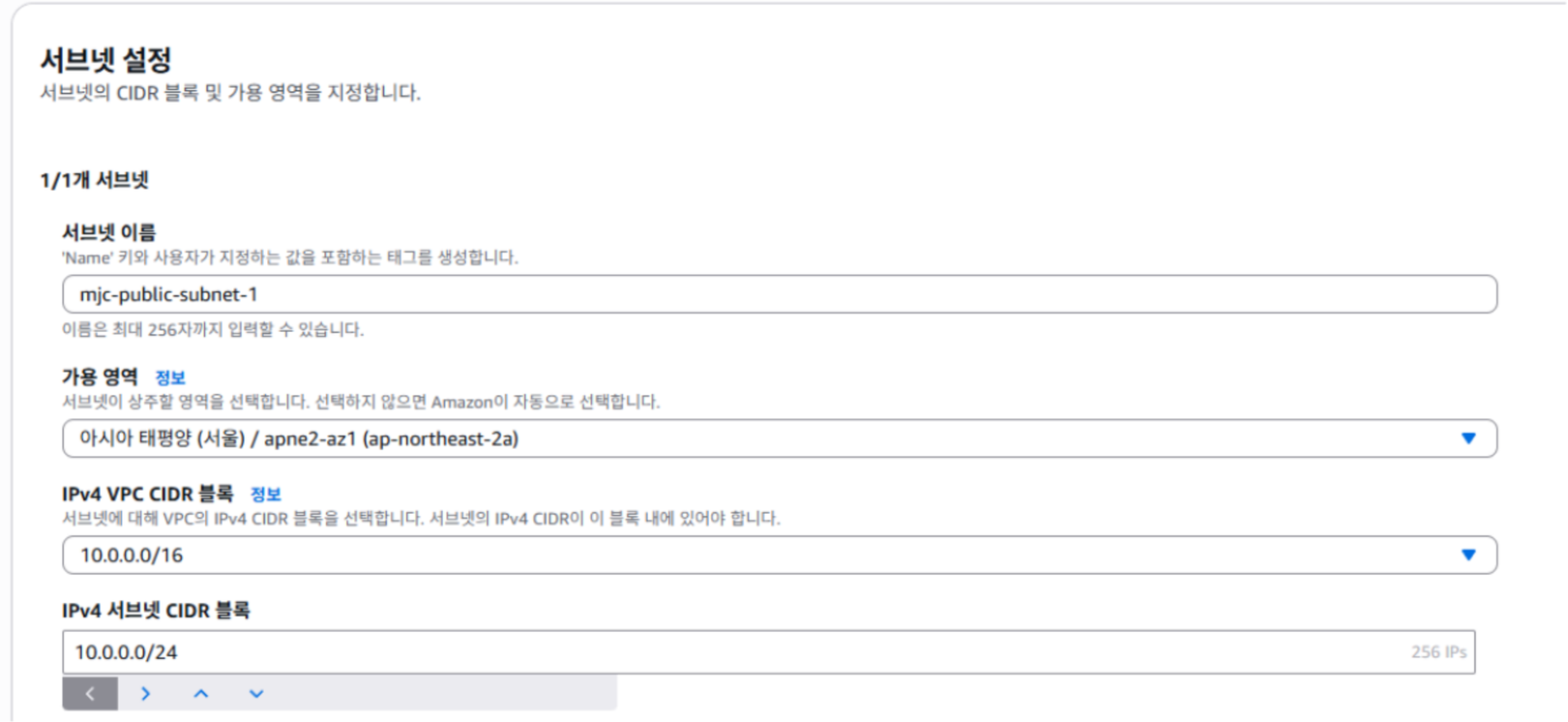
이름: mjc-vpc

IPv4 CIDR 블록: 10.0.0.0/16

3. VPC

3.2 Subnet 생성

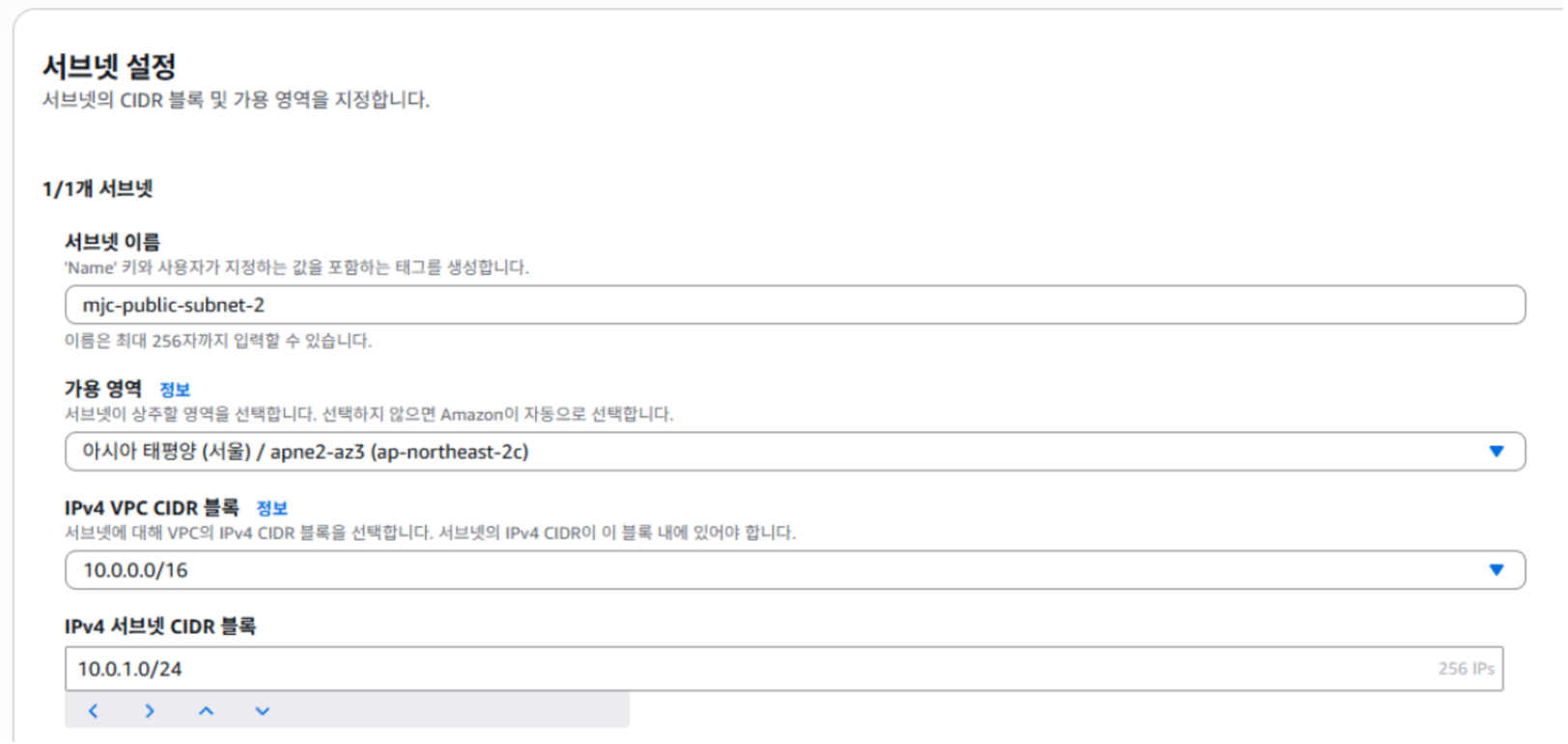
- mjc-public-subnet-2: ap-northeast-2a(10.0.0.0/24)



3. VPC

3.2 Subnet 생성

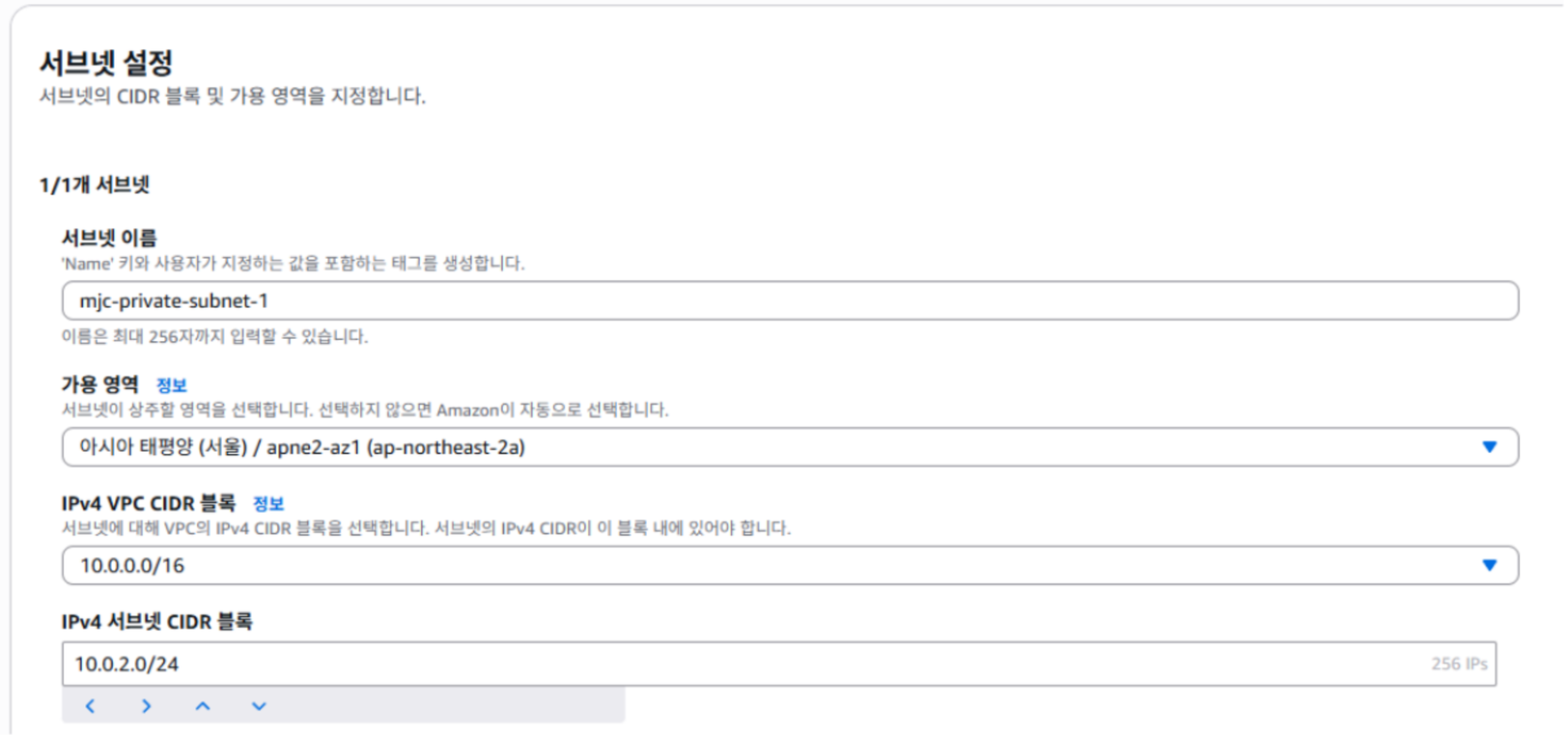
- mjc-public-subnet-2: ap-northeast-2c(10.0.1.0/24)



3. VPC

3.2 Subnet 생성

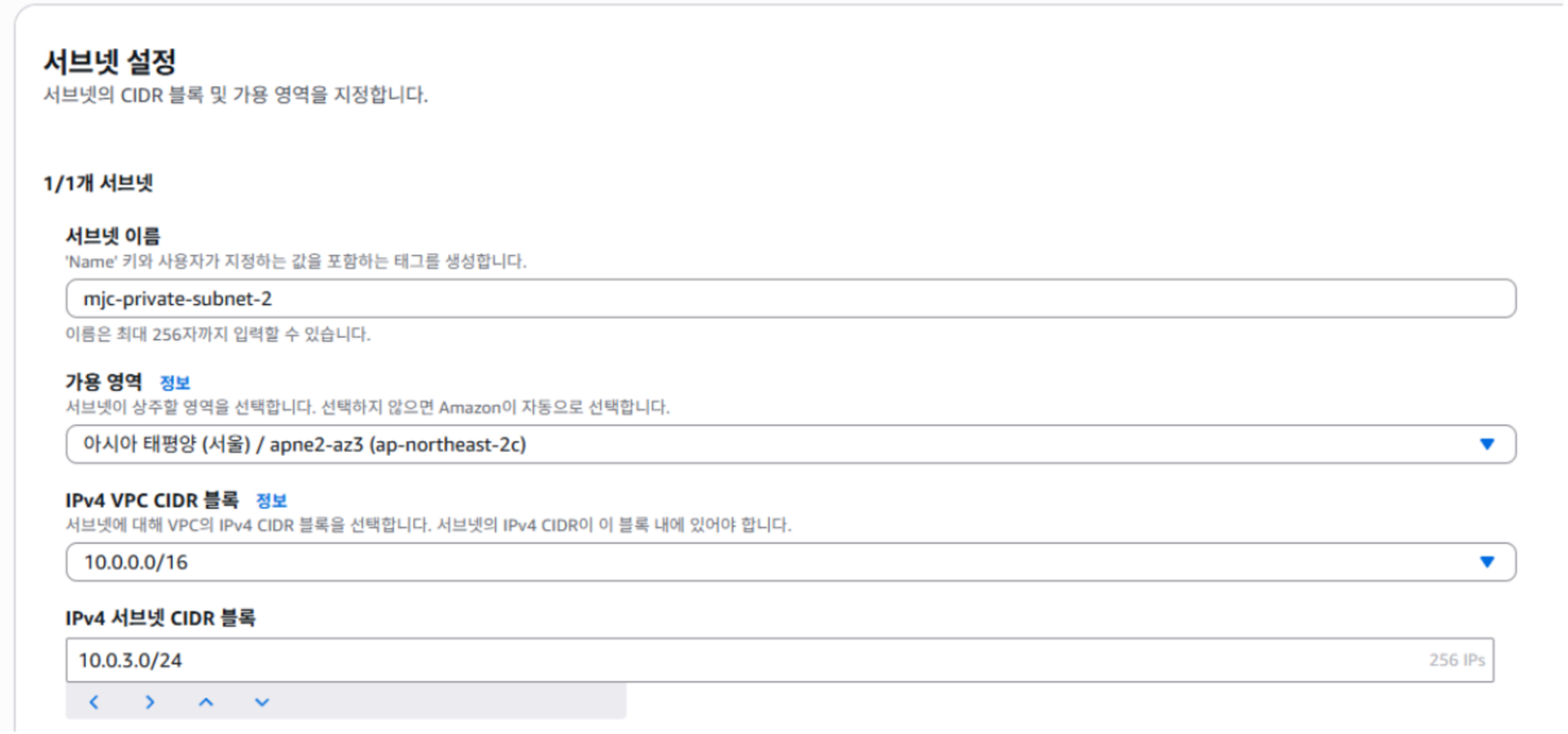
- mjc-private-subnet-2: ap-northeast-2a(10.0.2.0/24)



3. VPC

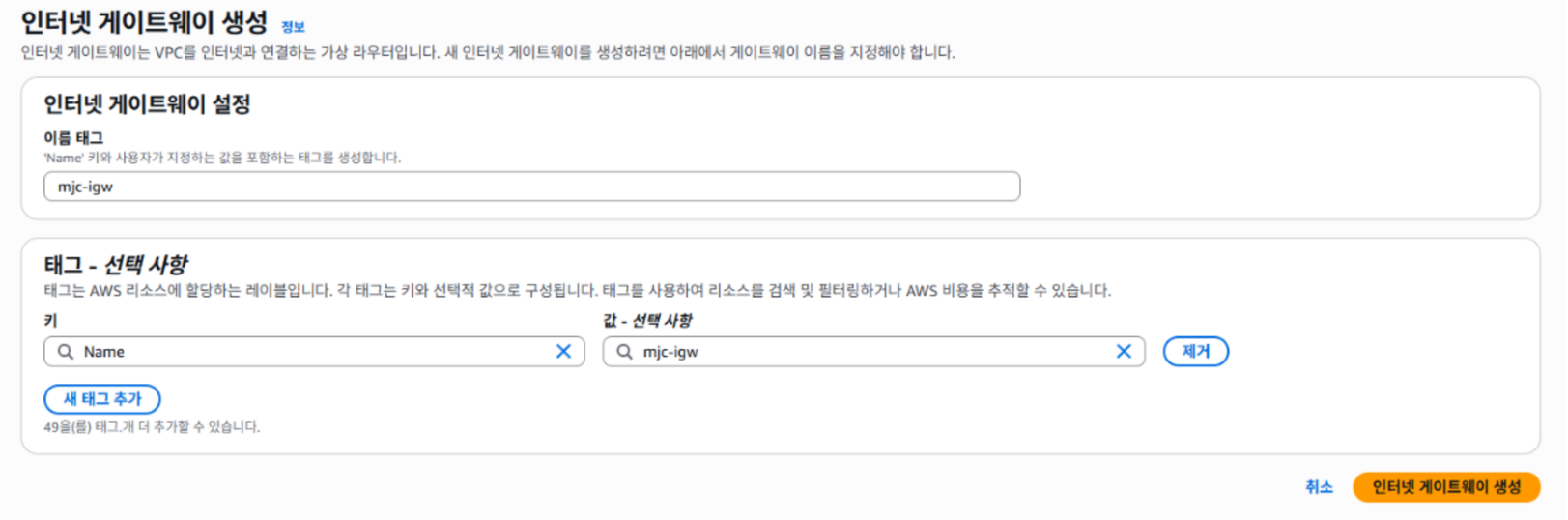
3.2 Subnet 생성

- mjc-private-subnet-2: ap-northeast-2c(10.0.3.0/24)

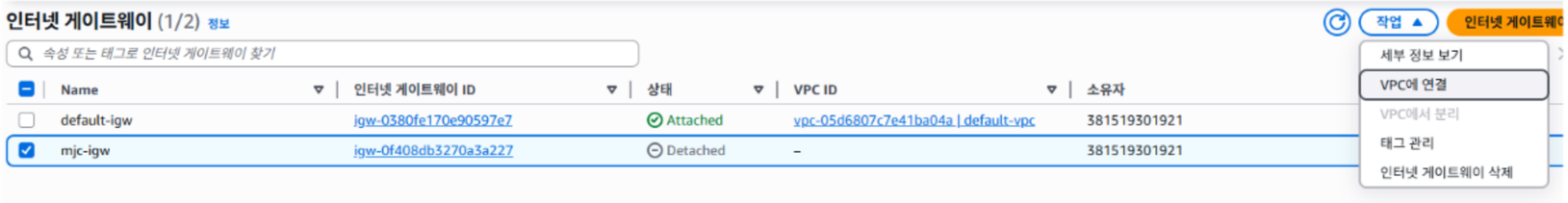


3. VPC

3.3 Internet Gateway 생성

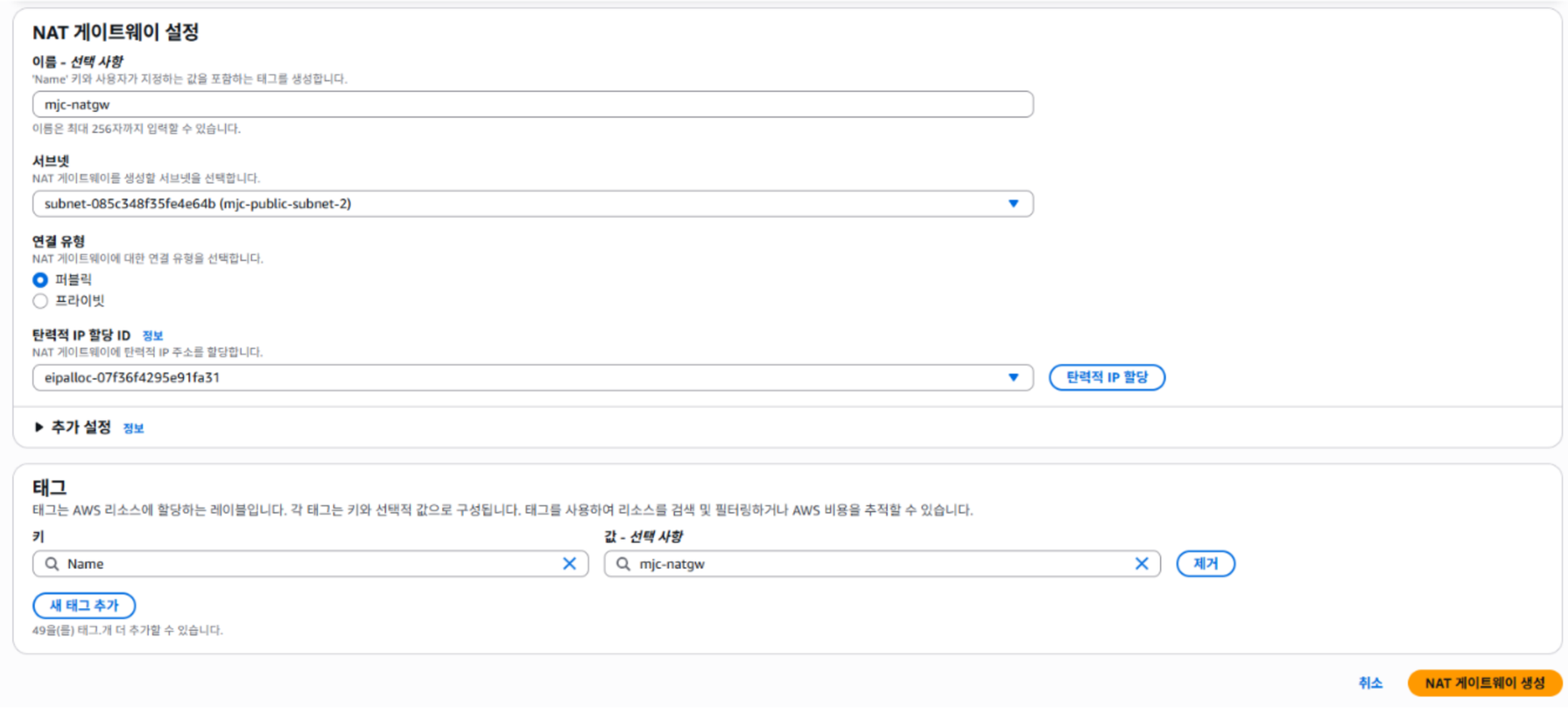


- VPC 연결



3. VPC

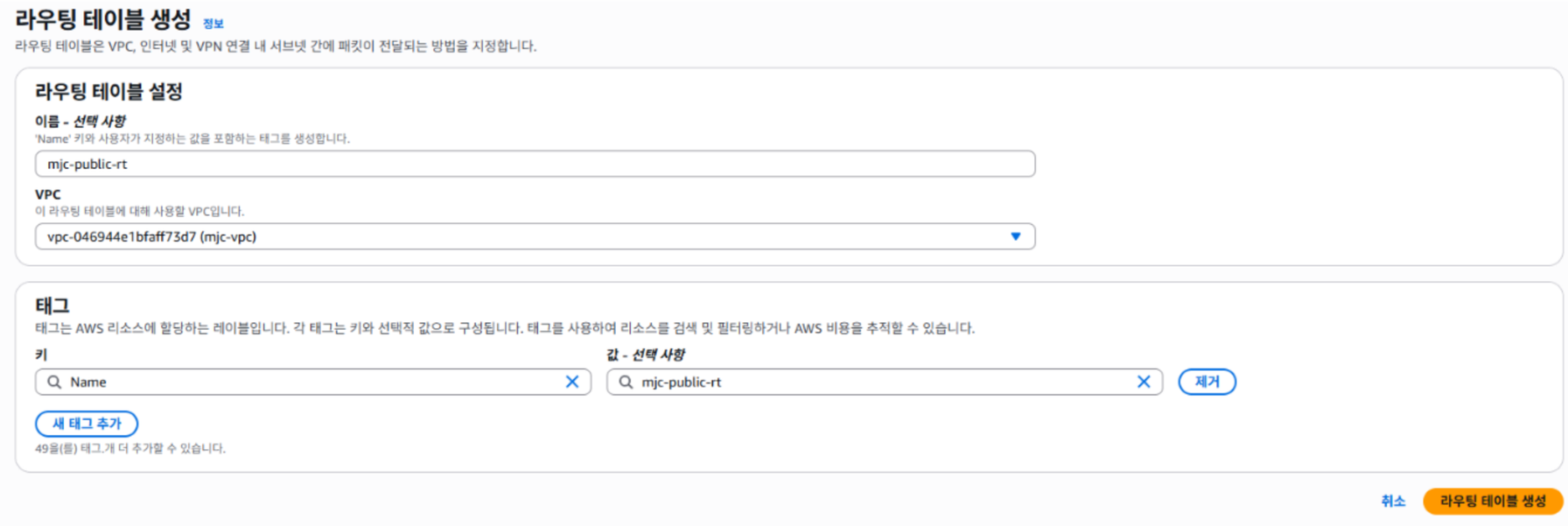
3.4 NAT Gateway 생성



3. VPC

3.5 Routing Table 생성

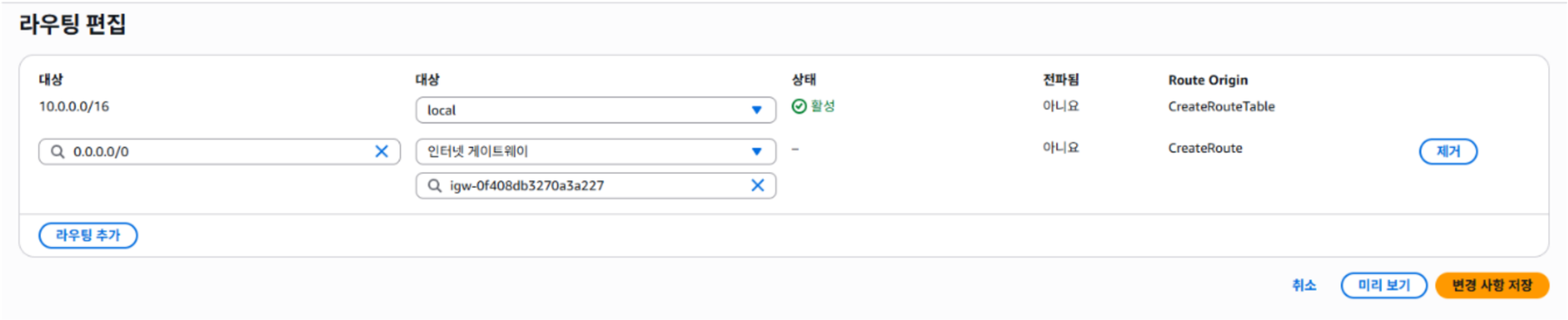
- **퍼블릭 전용 라우팅 테이블 생성**



3. VPC

3.5 Routing Table 생성

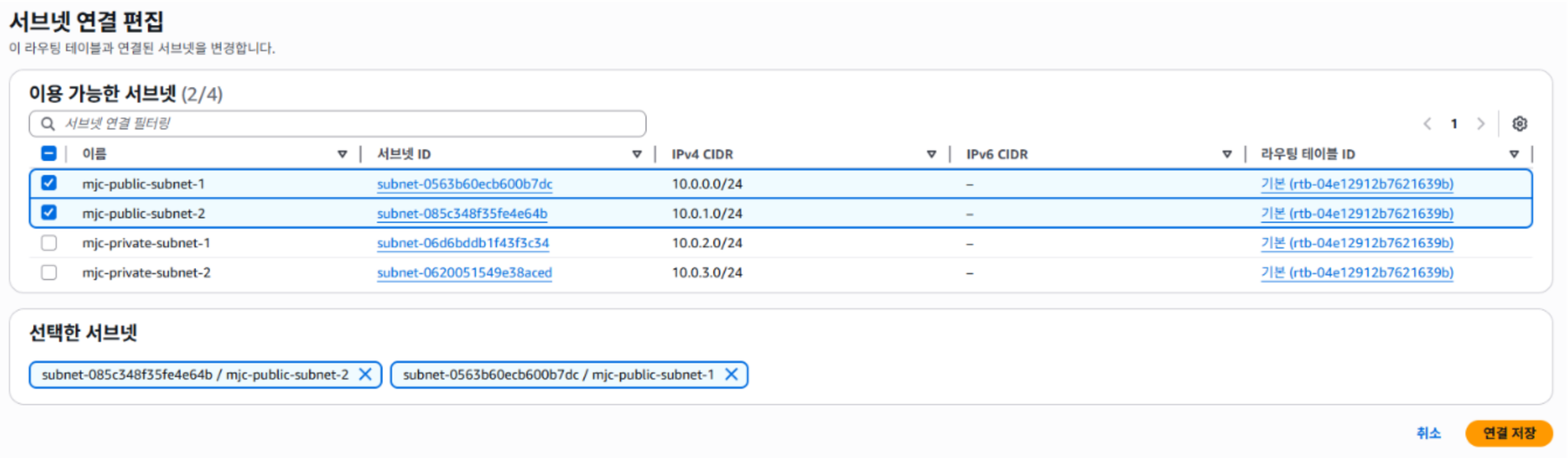
- 라우팅 편집



3. VPC

3.5 Routing Table 생성

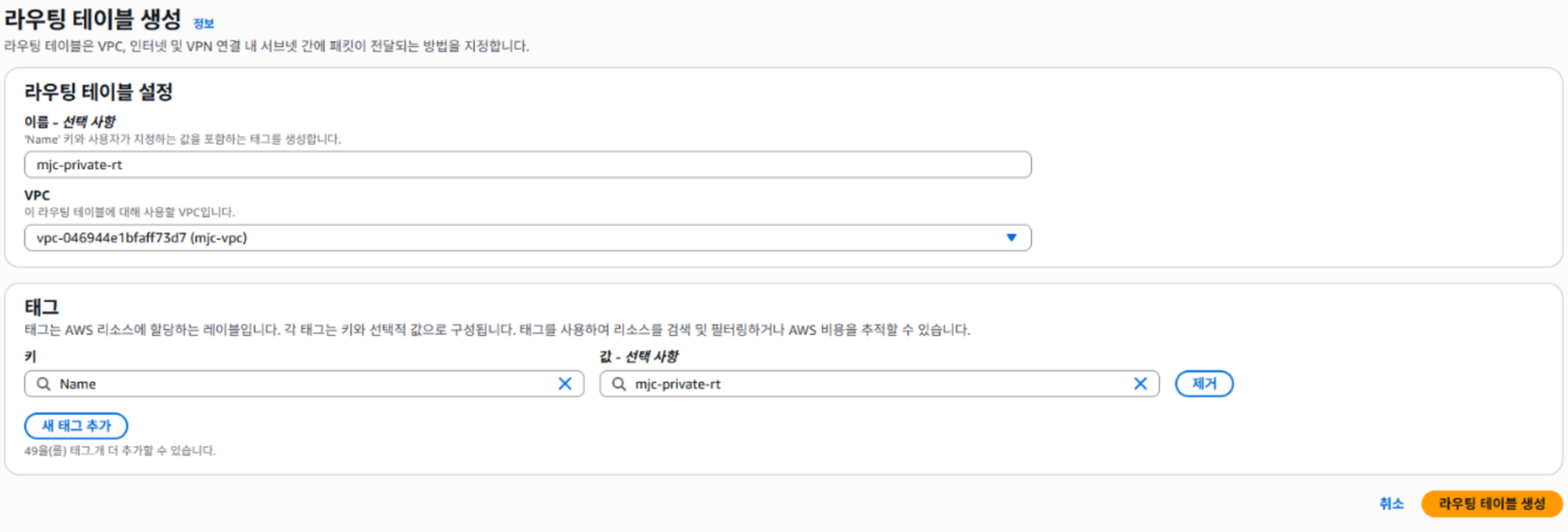
- **서브넷 연결 편집**



3. VPC

3.5 Routing Table 생성

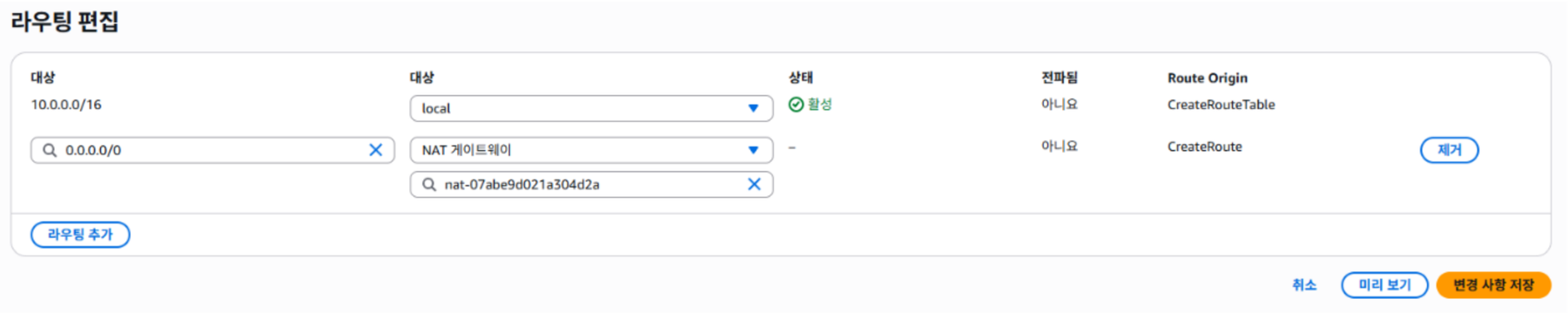
- **프라이빗 전용 라우팅 테이블 생성**



3. VPC

3.5 Routing Table 생성

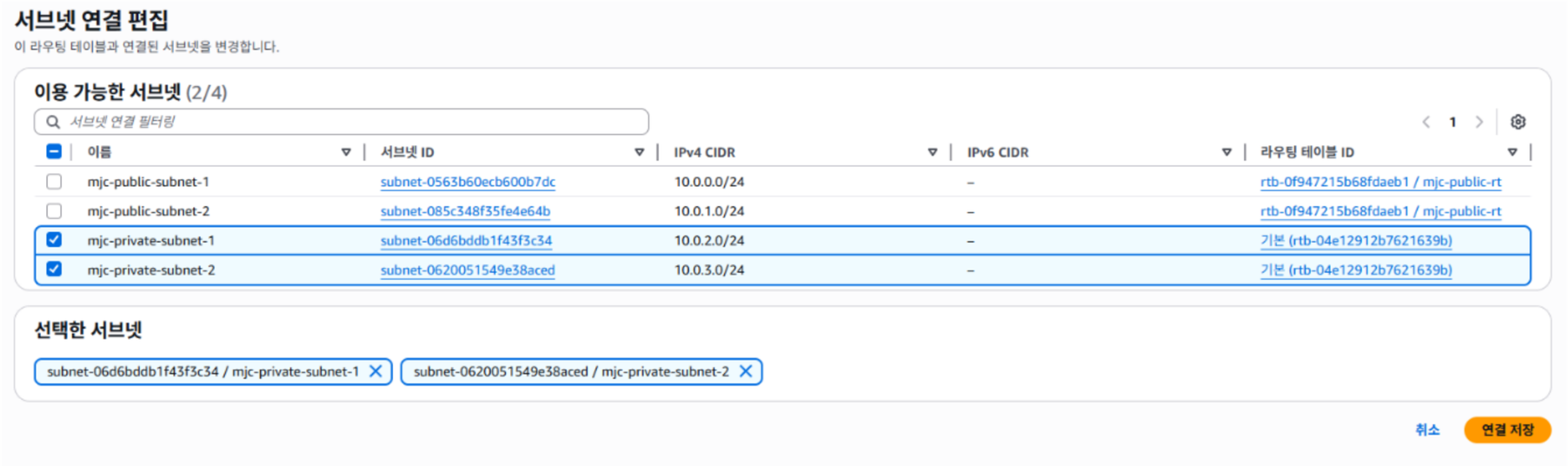
- 라우팅 편집



3. VPC

3.5 Routing Table 생성

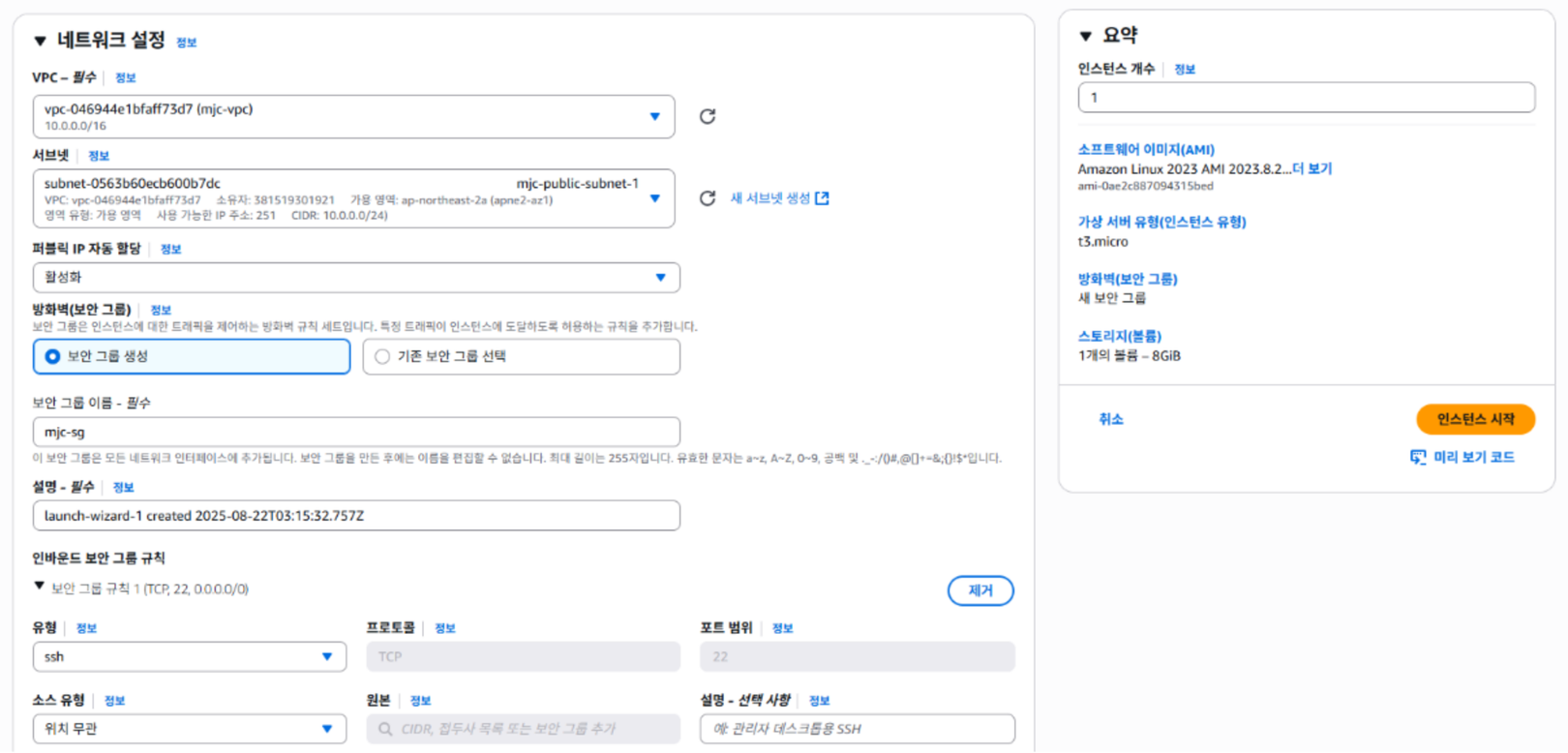
- **서브넷 연결 편집**



3. VPC

3.6 Bastion Host 생성 및 Instance 생성

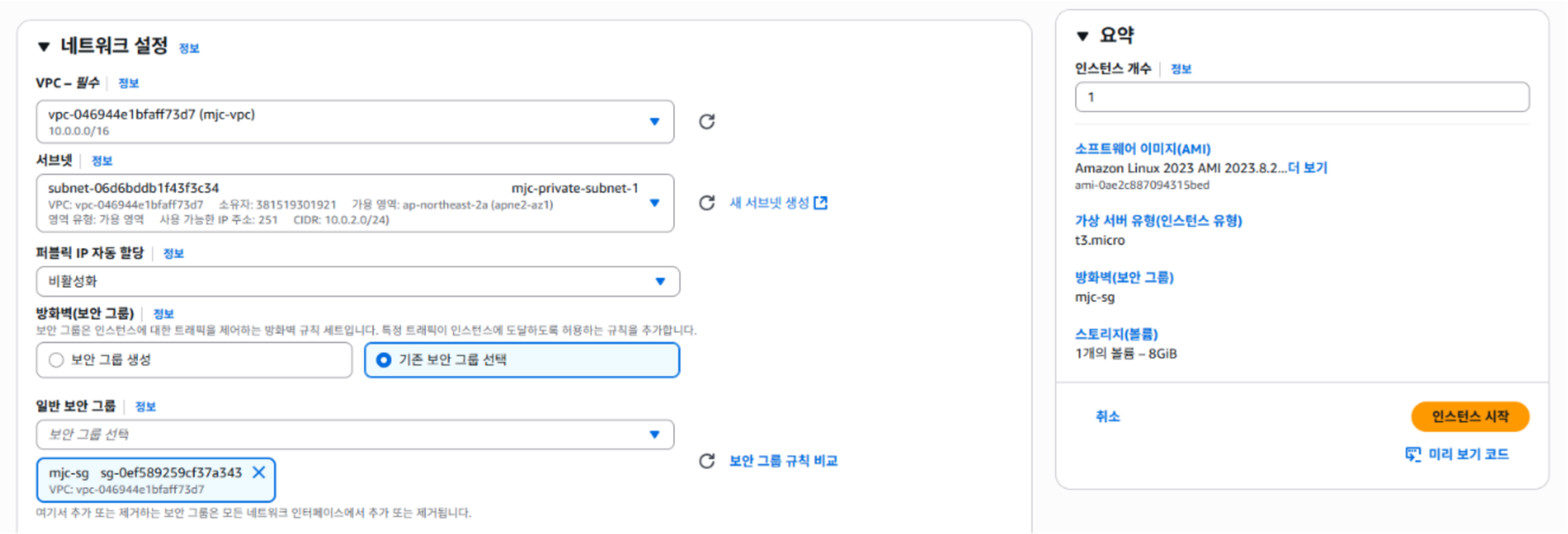
- mjc-bastion-inst



3. VPC

3.6 Bastion Host 생성 및 Instance 생성

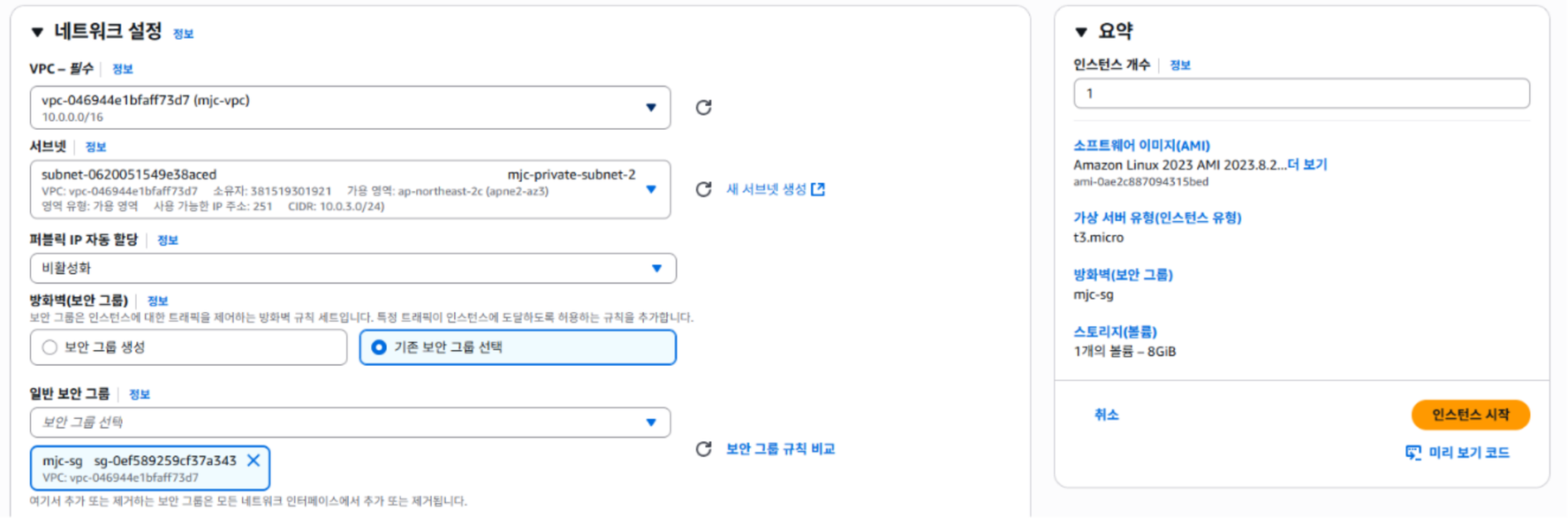
- mjc-private-inst 1



3. VPC

3.6 Bastion Host 생성 및 Instance 생성

- mjc-private-inst 2



3. VPC

3.7 접속

- ssh를 이용하여 bastion host에 접속

```
Connecting to 13.124.232.208:22...
Connection established.
To escape to local shell, press 'Ctrl+Alt+]'.

WARNING! The remote SSH server rejected X11 forwarding request.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-0-147 ~]$
[ec2-user@ip-10-0-0-147 ~]$
```

3. VPC

3.7 접속

- keypair.pem 파일 생성 -> vi keypair.pem
- chmod 400 kpair.pem 입력


```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEA2VujpL0wm0CDr4llB6EMlAfKCmzt37mECSdZdacHGikosknb
uLLxdzFD7ePxzMimlNfUxr/QGgqe5olT1vIMyT7c6z5yYcEWyYmG+v/VaUcnXqAr
RLn6cmDpGai2OahHjDy3Hr2QRJEMIcDqXgTc2dTqJskLDZRec/m28bMH4gIFSfPx
tZzpilhia70zkZpaaKF3WLqX+7XVStd8+nQjycsBHQdjp+dbHMDNO7yH0oYOb7TJ
lNmYe1E+YfOItCu0Iqb/S8EpIBg4MbD2x0qZ2Y1B/MZYw8MNDlDawO2yOV6QK0wU
uAu9fzCvVAwCgs9Ogw4yLY2RHqB1FOcgVzj+lQIDAQABAoIBAQCSd2AEaMBSPdz+
8nSz+WBCeCU7dT+t3JMOoAd2coEkrcuQqJh5E4wHJ5mAunN77VHjRrVI4IRYqAN/
kdZ/J0habBPuKXxmRvkWAaiFhtPGR35OE1xUyYXsgUVSVw/ENpNP+rHi8xY+PgeQ
zzCe/mPcHc9TJ3JdSjeDqygOIvgS4M+4gOVYxIHCkJk6jiahV5I63KaB8muC/Ned
/J48ZqS1wsm727Rd6+UFx+pPDLjcdPZ08E4X67O+tOUSWkMpybQ8VHYXp9YzStOQ
T3XZcbFZudgytYE+LPdWSPrlanYliPvI0rLQTcTpNDqECwgFotA4DTVBPKJ2BU1W
VX4XcVNBAoGBAPDaiXBOmwI3QQH6W0z8Z5x+A4FZAJb2HajMvauMxWYBzGiml FWq
MwKlifqazeB2MaMjHmpYmgW2HJxVVNWNRP5kdmkrFVTXxlNziEeDGwmCxkz/GikL
4DcX0kFVBPAxfR60zUNCvHC9MS9yUlg2zwDDZk3JQoBbxTxR6emD8OGdAoGBAOcG
2Vl2w5uUrXg876xFAOoqwn8LWYNQplxnicblPjrXAuBlkNWollUvwcLJPMuwO/Z4
YkaQDHJ3zXBf75pyDtjcJ8jNi4HDhRwDzsxEu70GPZV9ll4MahQwwzQu6DE/u4BE
5ROYKC/0FnTmo8Lt+LvfOo5jJWY8z8VZRe1fuxtZAoGASOUrcaUovvvwHeQi2+MM
jM2KzAwe5JGqmlsVEKWzXNMwhX/whdjURXnKj+FobAX5sfFAFFmqjA35DMu6uENg
XUoUmHHkq9/+c2VlGpewiClfURBoX4Hro2D/TqynqPahcriv4yqAk/J+v2hvlyRH
dGivPt2p1eVwhXgZPoajUfUCgYEAmkYkmRlZ/mcGgCiRFETwG296ekQl/l64+TPa
+YrUR46VSSwJz+PzXOeLSHwBsTN6llhEqEJBTZ9a/RPTeRt5dq9tkdZQttLN7KtO
SXIMvGmSbmVzKFK588msUQiR3XFl0H4xSyhsjbk91cq+yMkJNMwY2kSzENbDGFJJ
M/6df+ECgYEA4E3LEqpGCt4Lu7WY/vXYXjY9HgrSF1vS+Nf6mcviDkb7jg5ehAVW
4Tq3/4o58sBU6D+fUUfN+XyxIsCIuvvLk0hvHkXd+a30Rsy6FYTXfctj9qdfdw1S
YdEWUHvbbdD6LctlouUNVZcoTm3gANynZ6yZt9XeXsdeeFwhYnCg7ws=
-----END RSA PRIVATE KEY----
```

3. VPC

3.7 접속

- mjc-private-inst1에 접속 -> ssh -i "keypair.pem" ec2-user@주소

```
[ec2-user@ip-10-0-0-147 ~]$ ssh -i "keypair.pem" ec2-user@10.0.2.58
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-10-0-2-58 ~]$
```

- 외부 연결 확인

```
[ec2-user@ip-10-0-2-58 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
```

3. VPC

3.7 접속

- mjc-private-inst2에 접속

```
[ec2-user@ip-10-0-0-147 ~]$ ssh -i "keypair.pem" ec2-user@10.0.3.238
The authenticity of host '10.0.3.238 (10.0.3.238)' can't be established.
ED25519 key fingerprint is SHA256:dIfcSmwbCO0ovROzG/UZ9SlZWKgnAl7s1tDC0L4keFo.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.3.238' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-3-238 ~]$
```

- 외부 연결 확인

```
[ec2-user@ip-10-0-3-238 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
```